RISMS – System Test Report

1. Introduction

This document specifies the black-box test cases for the Retail Inventory and Stock Management System (RISMS).

Scope. All console menu operations: login, product add/update/remove, search, listing, stock in/out, stock check, stock history, inventory summary, and file persistence. Out of scope: performance load testing and multi-user concurrency.

All 61 cases are scripted in `tests/system_tests.sh`.

Base test data.

Product ID	SKU	Name	Category	Unit price (₹)	Threshold
P0001	SKU-001	Rice 5kg	Grocery	450.00	5
P0002	SKU-002	Soap	(blank)	10.00	0

2. Authentication (UC01, FR01, SEC-01)

All 8 authentication cases passed.

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-AUTH-01	FR01, UC01	Default account seeded on first run	Start app with no `users.txt`	`[INFO] No users found - seeded default account (admin / Admin@123).` and login prompt shown	Pass
TC-AUTH-02	FR01, SEC-01	Valid login	`admin` / `Admin@123`	`[OK] Welcome, admin!`, main menu with options 0–10 shown	Pass
TC-AUTH-03	SEC-01	Wrong password, attempts counted	`admin` / `wrong` (×2)	`Incorrect password. 2 attempt(s) remaining.` then `1 attempt(s) remaining.`	Pass
TC-AUTH-04	SEC-01	Lockout after 3 failed attempts	`admin` / `wrong` (×3)	`Too many failed attempts. Account locked for 300 second(s).`	Pass
TC-AUTH-05	SEC-01	Correct password during lockout	After TC-AUTH-04, `admin` / `Admin@123`	`Account locked. Try again in N second(s).`, no menu access	Pass

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-AUTH-06	SEC-01	Unknown username	`ghost` / `x`	Generic `Incorrect username or password.` (does not reveal whether user exists)	Pass
TC-AUTH-07	UC01	Quit at login	`quit`	`[INFO] Goodbye.`, program exits with no menu shown	Pass
TC-AUTH-08	SEC-01, BR09	Lockout persists across restart	Trigger lockout, restart app, log in with correct password	Still locked until 300 s window expires	Pass

3. Product Management (UC02–UC04, FR02–FR04)

All 17 cases passed.

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-PRD-01	FR02, UC02	Add valid product	Option 1 → SKU `SKU-001`, Name `Rice 5kg`, Category `Grocery`, Price `450`, Threshold `5`	`[OK] Product added with id P0001.` Detail shows qty 0, status LOW STOCK	Pass
TC-PRD-02	FR02	Add with optional category blank	SKU `SKU-002`, Name `Soap`, Category (Enter), Price `10`, Threshold `0`	Added as P0002, category shown as `(none)`	Pass
TC-PRD-03	VFR01, BR01	Duplicate SKU rejected	Add again with SKU `SKU-001`	`A product with sku 'SKU-001' already exists.` No record added	Pass
TC-PRD-04	VFR02, BR02	Empty SKU	SKU (Enter)	`SKU cannot be empty.`, re-prompt	Pass
TC-PRD-05	VFR02, BR02	Empty name	Name (Enter)	`Name cannot be empty.`, re-prompt	Pass
TC-PRD-06	VFR02, BR03	Negative unit price	Price `-5`	`Unit price cannot be negative.` Product not added	Pass
TC-PRD-07	VFR02, NFR02	Non-numeric price	Price `abc`	`Enter a valid number.`, re-prompt	Pass
TC-PRD-08	VFR02, NFR02	Non-integer threshold	Threshold `12abc`	`Enter a valid whole number.`, re-prompt	Pass
TC-PRD-09	VFR02, BR03	Negative threshold	Threshold `-2`	`Low-stock threshold cannot be negative.` Product not added	Pass

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-PRD-10	VFR02	Delimiter in name	Name `Bad|Name`	`Product name cannot contain '|' or newlines.`	Pass
TC-PRD-11	FR03, UC03	Update fields, keep others	Option 2 → `P0001`, Enter on SKU, Name `Basmati Rice 5kg`, Enter, Price `500`, Enter	`[OK] Product P0001 updated.` Name and price changed; SKU, category, threshold, qty unchanged	Pass
TC-PRD-12	FR03, BR01	Update to another product's SKU	Update P0001 with SKU `SKU-002`	`SKU 'SKU-002' is already used by product P0002`	Pass
TC-PRD-13	FR03	Update non-existent product	Product id `P0099`	`No product with id: P0099`	Pass
TC-PRD-14	FR04, UC04	Remove with confirmation	Option 3 → `P0002` → `y`	Detail shown, then `[OK] Product P0002 removed.` Not in listing	Pass
TC-PRD-15	FR04, UC04	Cancel removal	Option 3 → `P0002` → `n`	`[INFO] Cancelled.` Product still listed	Pass
TC-PRD-16	FR04, NFR02	Invalid confirmation answer	`maybe` at y/n prompt	`Please answer y or n.`, re-prompt	Pass
TC-PRD-17	FR04	Remove non-existent product	Product id `P0099`	`No product with id: P0099`	Pass

4. Search, View and Listing (UC05, UC06, UC10, FR05, FR06, FR11)

All 7 cases passed.

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-SRC-01	FR05, VFR03, BR04	Exact name match	Option 4 → `Rice 5kg`	Row `P0001 SKU-001 Rice 5kg qty=… price=450.00`	Pass
TC-SRC-02	FR05	Case-insensitive search	Option 4 → `RICE`	P0001 returned	Pass
TC-SRC-03	FR05	Partial match	Option 4 → `5k`	P0001 returned	Pass
TC-SRC-04	VFR04, BR05	No match	Option 4 → `xyz`	`[INFO] No products matched "xyz".`	Pass
TC-SRC-05	FR06, UC06	View product details	Add or update any product	Detail block shows ID, SKU, Name, Category, Unit Price, Quantity, Threshold, Status	Pass

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-SRC-06	FR11, UC10	List all products	Option 9 with P0001, P0002 stored	One row per product; low-stock rows tagged `[LOW STOCK]`	Pass
TC-SRC-07	FR11, UC10	List with empty inventory	Option 9 on fresh data	`[INFO] No products recorded yet.`	Pass

5. Stock Adjustment, Availability and History (UC07–UC09, FR07–FR10)

All 14 cases passed.

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-STK-01	FR07, VFR05, BR06	Stock-in increases quantity	Option 5 → `P0001`, qty `20`, reason `Restock` (qty was 0)	`[OK] Stock added successfully.` `New quantity: 20`	Pass
TC-STK-02	FR09, VFR07, BR08	Stock-in logged	Inspect `ledger.dat` after TC-STK-01	`M000001|P0001|IN|20|20|<timestamp>|admin|Restock`	Pass
TC-STK-03	FR07, NFR02	Stock-in qty 0 / negative	Qty `0`, then `-3`	`Enter a number between 1 and 1000000000.`, re-prompt	Pass
TC-STK-04	FR07, NFR02	Stock-in non-numeric qty	Qty `abc`	`Enter a valid whole number.`, re-prompt	Pass
TC-STK-05	FR07	Stock-in unknown product	`P0099`, qty `4`, reason `x`	`No product with id: P0099`; no ledger line	Pass
TC-STK-06	FR07, BR07	Stock-out within stock	Qty 20 → option 6, `P0001`, qty `15`, `Sale`	`Remaining quantity: 5`; ledger change `-15`	Pass
TC-STK-07	VFR06, BR07, SEC-03	Stock-out exceeds stock	Qty 20 → stock-out `50`	`Insufficient stock. Available quantity: 20`; qty unchanged, no ledger line	Pass
TC-STK-08	BR07	Stock-out exactly all stock (boundary)	Qty 10 → stock-out `10`	`Remaining quantity: 0`	Pass
TC-STK-09	BR07	Stock-out from zero stock	Qty 0 → stock-out `1`	`Insufficient stock. Available quantity: 0`	Pass
TC-STK-10	FR08, UC08	Check stock	Option 7 → `P0001`	`Current quantity: <qty>` matches last resulting quantity	Pass

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-STK-11	FR08	Check stock, unknown product	Option 7 → `P0099`	`No product with id: P0099`	Pass
TC-STK-12	FR10, UC09	View history	Option 8 → `P0001` after TC-STK-01 and TC-STK-06	Two rows, newest first: `M000002 | OUT | -15 | 5`, then `M000001 | IN | 20 | 20`, each with user and reason	Pass
TC-STK-13	FR10	History with no movements	Option 8 on a product never adjusted	`[INFO] No stock movements found.`	Pass
TC-STK-14	FR10	History, unknown product	Option 8 → `P0099`	`No product with id: P0099`	Pass

6. Inventory Summary (UC11, FR12, VFR09, BR10)

4 of 5 cases passed; TC-SUM-04 fails (DEF-01).

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-SUM-01	FR12, UC11	Totals correct	P0001 qty 5 @ ₹450, threshold 5 → option 10	`Total products: 1`, `Total units: 5`, `Total inventory value: 2250.00`, `Low-stock products: 1`	Pass
TC-SUM-02	VFR09, BR10	Qty equal to threshold is low stock (boundary)	P0001 qty 5, threshold 5	Counted as low stock; listing shows `[LOW STOCK]`	Pass
TC-SUM-03	VFR09, BR10	Qty above threshold is not low stock	P0001 qty 12, threshold 5	Listing shows no `[LOW STOCK]` tag	Pass
TC-SUM-04	VFR09, BR10, NFR03	Summary and listing agree when threshold is 0	P0002 threshold 0, stock-in 5 → option 9, then option 10	Both report P0002 consistently (either low or not low)	**Fail** — listing shows P0002 OK; summary counts it low (`Low-stock products: 2`)
TC-SUM-05	FR12	Empty inventory	Option 10 on fresh data	All totals 0, `Total inventory value: 0.00`	Pass

7. Persistence, Validation and Robustness (FR13, FR14, NFR01–NFR08)

All 10 cases passed; TC-ROB-03 passes with a minor display defect (DEF-02).

TC ID	Req	Scenario	Input / steps	Expected result	Status
TC-PER-01	FR13, DR01	Products written to file	Add P0001, exit, open `data/products.txt`	`P0001|SKU-001|Rice 5kg|Grocery|450.00|0|5`	Pass
TC-PER-02	FR13, DR04	Ledger written to file	After a stock-in, open `data/ledger.dat`	One line per successful movement, IDs M000001, M000002…	Pass
TC-PER-03	FR14, VFR08, BR09, NFR04	Data reloaded after restart	Update + stock-in, exit, restart, options 7, 8, 9	Updated name/price, qty 12 and history row all still present	Pass
TC-PER-04	FR14, NFR08	First run with no data files	Delete `products.txt`, `ledger.dat`, `users.txt`; start app	Starts normally, empty inventory, files created on first write	Pass
TC-ROB-01	NFR03	Corrupt product line	Append `garbage line no pipes` to `products.txt`, restart, option 9	Valid products listed; bad line skipped; warning in `data/error.log`	Pass
TC-ROB-02	NFR03, SEC-04	Corrupt ledger line	Append `junk` to `ledger.dat`, restart, option 8	Valid history shown; bad line skipped	Pass
TC-ROB-03	SEC-04	Delimiter in stock reason	Stock-in with reason `bad|reason`	Ledger stays valid; file stores `bad/reason`	Pass (see DEF-02)
TC-VAL-01	NFR01, NFR02	Menu choice out of range	Main menu `11`, then `-1`	`Enter a number between 0 and 10.`, re-prompt	Pass
TC-VAL-02	NFR01, NFR02	Menu choice non-numeric	Main menu `abc`	`Enter a valid whole number.`, re-prompt	Pass
TC-VAL-03	NFR03	Input stream closed mid-operation	Pipe input ending in the middle of Add Product	`[INFO] Input closed. Goodbye.`, clean exit, no partial record	Pass

8. Requirements-to-Test Traceability

Requirement	Test cases	Result
FR01 User access	TC-AUTH-01 – TC-AUTH-07	Pass
FR02 Add product	TC-PRD-01, TC-PRD-02	Pass
FR03 Modify product	TC-PRD-11 – TC-PRD-13	Pass
FR04 Remove product	TC-PRD-14 – TC-PRD-17	Pass
FR05 Search product	TC-SRC-01 – TC-SRC-04	Pass (name only)
FR06 View product details	TC-SRC-05	Pass
FR07 Adjust stock	TC-STK-01, TC-STK-03 – TC-STK-09	Pass
FR08 Check stock availability	TC-STK-10, TC-STK-11	Pass
FR09 Record stock adjustments	TC-STK-02	Pass
FR10 View stock movement history	TC-STK-12 – TC-STK-14	Pass
FR11 View inventory	TC-SRC-06, TC-SRC-07	Pass
FR12 Inventory summary	TC-SUM-01 – TC-SUM-05	Fail (TC-SUM-04)
FR13 Store data in files	TC-PER-01, TC-PER-02	Pass
FR14 Load data on startup	TC-PER-03, TC-PER-04	Pass
VFR01 Duplicate ID rejected	TC-PRD-03	Pass (by SKU)
VFR02 Invalid input rejected	TC-PRD-04 – TC-PRD-10	Pass
VFR03 Valid search returns record	TC-SRC-01	Pass
VFR04 Unknown key → not found	TC-SRC-04	Pass
VFR05 Stock-in calculation	TC-STK-01	Pass
VFR06 Stock-out ⩽ available	TC-STK-07 – TC-STK-09	Pass
VFR07 Stock changes logged	TC-STK-02	Pass
VFR08 Data survives restart	TC-PER-03	Pass
VFR09 Low-stock threshold	TC-SUM-02 – TC-SUM-04	Fail (TC-SUM-04)

Requirement	Test cases	Result
SEC-01 Authorized access only	TC-AUTH-02 – TC-AUTH-06, TC-AUTH-08	Pass
SEC-02 Restrict modification/removal	TC-AUTH-02, TC-PRD-14, TC-PRD-15	Partial (DEV-03)
SEC-03 Validate stock adjustments	TC-STK-03, TC-STK-04, TC-STK-07	Pass
SEC-04 History integrity	TC-ROB-02, TC-ROB-03	Pass

9. Execution Summary, Defects and SRS Deviations

60 of 61 test cases passed (98.4%); one functional defect and one minor display defect were found,.

Module	Cases	Pass	Fail
Authentication	8	8	0
Product management	17	17	0
Search, view, listing	7	7	0
Stock adjustment and history	14	14	0
Inventory summary	5	4	1
Persistence, validation, robustness	10	10	0
Total	**61**	**60**	**1**

Defects

ID	Severity	Found by	Description	Suggested fix
DEF-01	Medium	TC-SUM-04	Low-stock rule differs between listing and summary for products with threshold 0. `Product::isLowStock()` uses the product's threshold (0); `ReportService::inventorySummary()` falls back to the default 10. A product with qty 5, threshold 0 shows OK in the list but is counted low in the summary.	Use one rule in both places, e.g. move the fallback into `Product::isLowStock(int defaultThreshold)` and call it from both.

ID	Severity	Found by	Description	Suggested fix
DEF-02	Low	TC-ROB-03	A `|` in a stock reason is saved as `/` in `ledger.dat`, but the in-memory copy keeps `|`, so history shows `bad|reason` until restart and `bad/reason` after.	Sanitise the reason in `StockService` before building the movement, or reject `|` like product fields do.